

UCampus Cafeteria Management System Assignment

Problem Statement

UCampus has one central cafeteria that serves students and staff. Every food item belongs to the same cafeteria, and the cafeteria has a common service charge that applies to all food items. Develop a Java program to manage the cafeteria menu.

Create a class named FoodItem.

Instance Variables

- int itemId
- String itemName
- double price

Static Variables

- String cafeteriaName = "UCampus Cafeteria"
- double serviceCharge = 10.0
- int totalFoodItems = 0

Instance Methods

1. addFoodItem(int id, String name, double price)
 - Assign values to the instance variables.
 - Increase totalFoodItems by 1 whenever a new food item is added.
2. calculateFinalPrice()
 - $\text{Final Price} = \text{Price} + (\text{Price} \times \text{Service Charge} / 100)$
3. displayItemDetails()
 - Display Item ID, Item Name, Original Price, and Final Price.

Static Methods

1. changeServiceCharge(double newCharge)
 - Update the service charge for the cafeteria.
2. displayCafeteriaDetails()
 - Display Cafeteria Name, Current Service Charge, and Total Food Items.

Sample Driver Code

```
FoodItem item1 = new FoodItem();
FoodItem item2 = new FoodItem();
FoodItem item3 = new FoodItem();

item1.addFoodItem(101, "Veg Sandwich", 80);
item2.addFoodItem(102, "Cold Coffee", 120);
item3.addFoodItem(103, "Paneer Wrap", 150);

FoodItem.displayCafeteriaDetails();

item1.displayItemDetails();
item2.displayItemDetails();
```

```
item3.displayItemDetails();  
  
FoodItem.changeServiceCharge(15);  
  
System.out.println("After Updating Service Charge");  
  
item1.displayItemDetails();  
item2.displayItemDetails();  
item3.displayItemDetails();
```